

ML Bias Detector

Fairness Audit Report

Model: logistic_regression_adult Dataset: adult_income_test Date: June 09, 2026

Audit Summary

This report presents the results of a fairness audit on the **logistic_regression_adult** model using the **adult_income_test** dataset. Sensitive attributes evaluated: **sex**, **race**. Any disparity exceeding **0.1** was flagged for review.

Attribute	Groups Evaluated	Flagged Disparities
sex	2	2
race	5	23

Key Findings

SEX: 2 disparity flag(s) across 2 groups. Largest gap — ACCURACY between Male and Female: 0.1195 (threshold 0.1).

RACE: 23 disparity flag(s) across 5 groups. Largest gap — ACCURACY between White and Other: 0.1228 (threshold 0.1).

Dataset Health Check

SEX: 1 issue(s) detected — label_distribution.

RACE: 6 issue(s) detected — imbalance_ratio, label_distribution, small_group.

These results are descriptive only. They show differences in model performance across groups but do not imply legal or causal conclusions.

DATASET AUDIT — DETAILED RESULTS

Attribute: sex

- [LABEL_DISTRIBUTION] Group 'Female' has a positive label rate of 11.9% vs overall rate of 24.9% (disparity: 0.1296).

Attribute: race

- [SMALL_GROUP] Group 'Amer-Indian-Eskimo' has only 64 records — metrics may be unreliable.
- [SMALL_GROUP] Group 'Other' has only 38 records — metrics may be unreliable.
- [IMBALANCE_RATIO] Group size imbalance ratio is 136.47x — the largest group is 136.47x bigger than the smallest.
- [LABEL_DISTRIBUTION] Group 'Other' has a positive label rate of 7.9% vs overall rate of 24.9% (disparity: 0.1701).
- [LABEL_DISTRIBUTION] Group 'Black' has a positive label rate of 14.3% vs overall rate of 24.9% (disparity: 0.1059).
- [LABEL_DISTRIBUTION] Group 'Amer-Indian-Eskimo' has a positive label rate of 9.4% vs overall rate of 24.9% (disparity: 0.1552).

MODEL AUDIT — SEX

Per-Group Metrics

Group	n	Accuracy	TPR	FPR	FNR	PPR	Precision	F1
Male	4065	0.7697	0.3978	0.0618	0.6022	0.1665	0.7445	0.5185
Female	1968	0.8892	0.3532	0.0381	0.6468	0.0757	0.557	0.4323

Confusion Matrix Values

Group	TP	TN	FP	FN
Male	504	2625	173	763
Female	83	1667	66	152

Flagged Disparities

Metric	Group A	Group B	Value A	Value B	Disparity	Threshold
ACCURACY	Male	Female	0.7697	0.8892	0.1195	0.1
PRECISION	Male	Female	0.7445	0.557	0.1875	0.1

MODEL AUDIT — RACE

Per-Group Metrics

Group	n	Accuracy	TPR	FPR	FNR	PPR	Precision	F1
White	5186	0.7983	0.3952	0.0571	0.6048	0.1464	0.7128	0.5085
Other	38	0.9211	0.3333	0.0286	0.6667	0.0526	0.5	0.4
Black	559	0.8766	0.2875	0.0251	0.7125	0.0626	0.6571	0.4
Amer-Indian-Eskimo	64	0.9219	0.3333	0.0172	0.6667	0.0469	0.6667	0.4444
Asian-Pac-Islander	186	0.8333	0.4545	0.0493	0.5455	0.1452	0.7407	0.5634

Confusion Matrix Values

Group	TP	TN	FP	FN
White	541	3599	218	828
Other	1	34	1	2
Black	23	467	12	57
Amer-Indian-Eskimo	2	57	1	4
Asian-Pac-Islander	20	135	7	24

Flagged Disparities

Metric	Group A	Group B	Value A	Value B	Disparity	Threshold
ACCURACY	White	Other	0.7983	0.9211	0.1228	0.1
PRECISION	White	Other	0.7128	0.5	0.2128	0.1
F1	White	Other	0.5085	0.4	0.1085	0.1
TPR	White	Black	0.3952	0.2875	0.1077	0.1
FNR	White	Black	0.6048	0.7125	0.1077	0.1
F1	White	Black	0.5085	0.4	0.1085	0.1
EQUALIZED_ODDS	White	Black	—	—	0.1077	0.1
ACCURACY	White	Amer-Indian-Eskimo	0.7983	0.9219	0.1236	0.1
PRECISION	Other	Black	0.5	0.6571	0.1571	0.1
PRECISION	Other	Amer-Indian-Eskimo	0.5	0.6667	0.1667	0.1
TPR	Other	Asian-Pac-Islander	0.3333	0.4545	0.1212	0.1
FNR	Other	Asian-Pac-Islander	0.6667	0.5455	0.1212	0.1
PRECISION	Other	Asian-Pac-Islander	0.5	0.7407	0.2407	0.1
F1	Other	Asian-Pac-Islander	0.4	0.5634	0.1634	0.1
EQUALIZED_ODDS	Other	Asian-Pac-Islander	—	—	0.1212	0.1

Metric	Group A	Group B	Value A	Value B	Disparity	Threshold
TPR	Black	Asian-Pac-Islander	0.2875	0.4545	0.167	0.1
FNR	Black	Asian-Pac-Islander	0.7125	0.5455	0.167	0.1
F1	Black	Asian-Pac-Islander	0.4	0.5634	0.1634	0.1
EQUALIZED_ODDS	Black	Asian-Pac-Islander	—	—	0.167	0.1
TPR	Amer-Indian-Eskimo	Asian-Pac-Islander	0.3333	0.4545	0.1212	0.1
FNR	Amer-Indian-Eskimo	Asian-Pac-Islander	0.6667	0.5455	0.1212	0.1
F1	Amer-Indian-Eskimo	Asian-Pac-Islander	0.4444	0.5634	0.119	0.1
EQUALIZED_ODDS	Amer-Indian-Eskimo	Asian-Pac-Islander	—	—	0.1212	0.1

AUDIT CONFIGURATION

The settings below were used for this audit. The same config will always produce the same report.

Setting	Value
Model Name	logistic_regression_adult
Model Path	models/model.joblib
Dataset Name	adult_income_test
Dataset Path	data/adult_test.csv
Sensitive Attributes	sex, race
Label Column	income
Positive Label	>50K
Threshold	0.1
Metrics Computed	accuracy, tpr, fpr, ppr, fnr, precision, f1, equalized_odds

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Generated by ml-bias-detector | June 09, 2026